

Philosophy 415: Question Set #2

Due on Friday, October 18, 2019

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1

In the “Preface” to Wittgenstein’s *Tractatus Logico-Philosophicus*, he introduces the aim of the book by saying: “The aim of the book is to draw a limit to thought, or rather – not to thought, but to the expression of thought: for in order to be able to draw a limit to thought, we should have to find both sides of the limit thinkable (i.e. we should have to be able to think what cannot be thought).” What kind of contradiction is this, according to Priest? How is its underlying structure similar to that of the set-theoretical paradoxes, for instance Russell’s paradox? *How is it different?*

Priest would call this a self-referential contradiction meaning that it is a set-theoretical paradox whose form fits that of the inclosure schema. The inclosure schema is noted as having the following form:

a. $\Omega = \{y \mid \phi(y)\}$ exists and $\psi(\Omega)$

b. *If $x \subset \Omega$ and $\psi(x)$, then :*

1. $\delta(x) \notin x$

and

2. $\delta(x) \in \Omega$

c. $\delta(\Omega) \notin \Omega$ and $\delta(\Omega) \in \Omega$

In English the Inclosure Scheme is read as: omega is the set of all y where phi of y exists, and psi of omega. If x is a subset of omega and psi of x , then delta of x such that it is not a member of x and delta of x such that it is a member of omega. The contradiction occurs when we have delta Ω such that it is not a member of Ω and delta of Ω is a member of Ω . Thus showing P and $\sim P$. This formalizes the structure of all self-referential contradictions including Russell’s paradox, the liar paradox, and the barber paradox.

If we apply the same inclosure schema structure to Wittgenstein’s sentence, our input values would be represented as such:

a. Ω = the set of all expressible thoughts

b. y = the set of all thinkable thoughts

c. ϕ = the property of being a thought

d. ψ = the property of existence

e. δ = diagonalize (an element in omega but not in x)

Meaning, that the set of all thoughts is the set of a thought such that the thought has the property of being a thought and exists, and the set of all thoughts exist (existence). If the set of thinkable thoughts is a subset of the set of all thoughts and the thought exists, then the derivative from the thought is that it is not a member of the set of thinkable thoughts (transcendence), and the derivative of thought is an element of the set of all thoughts (closure).

The difference is that by expressing the thoughts in Omega and then applying delta to it, leads to the invalidity by the Inclosure Schema.

2

On one of its versions, the Löwenhem-Skolem theorem establishes that, if ZFC set theory has any infinite model, it has infinitely many *countable* models. With reference to the idea of model theory, please explain in your own words what this means, how it threatens a kind of paradox, and how this seeming paradox can be resolved. Even if so, does the theorem still give us reason to doubt the truth of the ZFC axioms with respect to the reality they are supposed to describe? Explain why or why not.

First, model theory studies classes and structures and does so by examining semantical elements using syntactical elements. This is done because we want our set theoretical axioms to describe a well-defined arithmetic such that we can draw accurate and complete conclusions. Another way of saying this is that we want our modeled system to be sound and complete. Soundness just means that if B is derivable from A , A entails B . Completeness means that if A entails B , A is derivable from B . If we can prove both of these, then we have accurately and completely modeled a system that is both sound and complete.

Now knowing this, let's examine ZFC set theory. First, if ZFC set theory has any infinite model, then it has infinitely many countable models. In other words, since it has a model that is infinitely countable (or uncountable), it also has infinitely many models containing only countably many objects. This means that everything proven true using ZFC set theory (transfinite sets which led to the set-theoretical universe big V and uncountable sets), is also true for universes containing countably many sets. This weirdly shows that we cannot be certain that we are talking about anything bigger than a countable set.

This puts us in a paradoxical situation, where all results regarding transfinite cardinality like the cardinality of R ($|R|$, where R = Relevant Universe) is greater than the cardinality of R ($|R|$) is also true in a universe with only countable sets (ω). So, if we take a specified countable Model M , it will only have sets greater than ω .

We can resolve this paradox by standing outside of the particular model and instead view how it is functions in a bigger universe of sets. To prove this, we must reexamine what the paradox is saying. This paradox says that within the relevant universe, there exists two sets R and ω that don't have a one to one correspondence. However, let's say we are within Model M and we have both R and ω that are countable (property of being in M) and function the same as usual, are they isomorphic?

The answer is yes and no. If we are looking from a perspective outside of Model M , then yes, they are; however, if we are looking from within Model M , then no, a one-to-one correspondence is not itself in M . Even if this paradox can be resolved, the question still needs to be asked: "does the theorem still give us reason to doubt the truth of the ZFC axioms with respect to the reality they are supposed to describe?". The answer is yes we still have reason to doubt the validity of the ZFC axioms, because we still have not proved that they are both consistent and complete in regards to modeling "mathematical" reality which is the reality they are trying to describe.

3

In 1637, Pierre de Fermat famously made the following conjecture in the margin of a book (and said that he possessed a proof of it that was too short to write there): $a^n + b^n = c^n$ has no integer solutions a, b, c for any integral value of $n \geq 2$. After hundreds of years of work, the conjecture was finally proven by Andrew Wiles in 1995. Assuming Fermat did not actually possess a valid proof, discuss how a logicist, an intuitionist and a formalist would each understand the meaning and truth of the conjecture i) before 1637; ii) between 1637 and 1995; and iii) today. (Take the cartesian product of schools of thoughts and time periods)

Before 1637 the general understanding used to understand the meaning and truth of concepts like Fermat's conjecture was to use a proof of "construction". During this time, a proof of "construction" (usually by drawing) was used as a means to prove a proof in the tangible world that actually exists. For instance, if we used Euclidian geometry, then we would have to construct a proof such that we only used a straight edge, compass, and measurements that did not introduce shapes that were not made from the other two tools.

Moreover, during the 19th century, Kant believe that the answer should be grounded in perception of the thing (intuitionist). He believed that all mathematical judgments should be grounded in out intuition because counting is grounded in time and time is known through an intuition.

Futhermore, during the 19th and 20th century, the foundations debate began due to Frege and Cantor. These schools of thought each would have a different perspective on Fermat's conjecture.

First, Logicism in the late 19th century would argue that Fermat's conjecture could be reduced to only logic. We run into a problem with this because at the time we did not have a great explanation about what a number was.

Second, intuitionists during the 20th century would say that the answer can not be reduced to logic or some other formal system, but it's structure needs to be grounded in human experience or intuition.

Third, formalists in the 20th century would say that Fermat's problem requires a formal system to answer the conjecture.

Overall, the truth of the conjecture could only be answered in the confines of each of those schools of thought.

4

Explain how Berry's paradox (as in the notes for Week 3) can be fit into Priest's Inclosure Schema (see p. 134 of *Beyond the Limits of Thought* for the schema). (Hint: take $\delta(x)$ as the least number not in x , ψ as the property of being definable in 18 syllables or less, ϕ as the property of being definable in 30 syllables or less.)

First Berry's paradox is: "The least integer not nameable in fewer than nineteen syllables' is itself a name consisting of eighteen syllables; hence the least integer not nameable in fewer than nineteen syllables can be named in eighteen syllables, which is a contradiction". This paradox can fit into Priest's Inclosure Schema because it is a self-referential paradox. For instance, we can assign the variables of the sentence as:

1. $\Omega = n$ where n is a natural number definable in less than 19 syllables
2. $\psi(x)$ = is definable in less than 14 words
3. $\phi(x)$ = is a natural number definable in less than 19 syllables
4. $\delta(x)$ = least ordinal not in x

From this, we can see that Transcendence and Closure are satisfied from the following steps:

1. if x is definable in less than 14 words and $\delta(x) \notin (x)$ then:
2. $\delta(x) \in \Omega$
 - a. and this returns the contradiction:
3. $\delta(\Omega) \in \Omega$ and $\delta(\Omega) \notin \Omega$

Therefore, because this fits the form P and not P we can fit Berry's Paradox into Priest's Inclosure Schema.

5

According to the “iterative conception of a set,” as suggested by George Boolos in an article with that title, beginning with the empty set, sets are “formed” in ever-higher levels or stages. At each stage n , it is possible to form all possible sets of objects occurring at levels $n-1$ and lower (there is also said to be a stage ω , after all finite-numbered stages, etc.) Thus “sets are formed over and over again: ... a set is formed at every stage later than that at which it is first formed...” Accordingly “...there is no set of all sets; for every set is formed at some earliest stage, and has as members only individuals or sets formed at still earlier stages.” (Benacerraf and Putnam, *Philosophy of Mathematics*, pp. 492-93). Do you find this picture of the set-theoretical universe convincing? Why or why not? In your answer, please refer to i) what Priest calls the “Domain Principle” and ii) the consequences of Russell’s paradox

Boolos is painting a picture of a set-theoretical universe also known as “V” or the Transfinite Hierarchy. Boolos explanation shows how “V” is generated starting only from the empty set and culminating into infinity. I believe Boolos’ explanation of the set-theoretical universe is convincing.

First, Boolos explains that beginning with the empty set, we can build more sets from this set. This process is what is known as “diagonalization”. Diagonalization is the process of using an operation like the power set to diagonalize out of a set such that it creates a new set containing elements with the same property that is and is not a member of the totality. From this, we can use diagonalization to build larger sets.

For instance, if we start at the first stage, which is the empty set, then we can take the power set (diagonalizer) enough times such that we even reach stage $\mathbb{N}$ (the set of all natural numbers) and Omega (the first infinite ordinal set and the first infinite cardinal set). Once we have gotten to stage $\mathbb{N}$, we have proven a one-to-one correspondence with the set of natural numbers and the set of real numbers. Not only do we prove this, but we eventually prove the transfinite hierarchy of sets.

This however can lead to a contradiction when we think about the sets of all sets (Russell’s Paradox) and this contradiction must be resolved. However, let us first exam what is causing these reoccurring contradictions and the answer is the diagonalization function itself. Ironically, using this function to generate the transfinite hierarchy also generates the same underlying contradictions. Therefore, we are unable to use the diagonalization function to solve these contradictions pertaining the diagonalization function.

Some philosophers and mathematicians have offered resolutions to the problem. For instance, we could use Russel’s theory of types, or axioms restricting the set of all sets, or the iterative conception of sets. However, these do not provide an adequate solution to infinite totalities and just cover the problem of paradoxical structures.

Priest provides an illustration of how this is done with his Domain Principle. The Domain Principle is: every “potential infinity” presupposes an actual infinity. We say this because in order to think about a potential infinity we must first assume that in some way that infinity actually exists along with its successors. This is not to say that we are making up numbers or collections of elements out of thin air, that would be chaotic. Instead we are using both ordinary math and language to discuss variables that cover infinite domains, but first, we must assume that those domains exist.

If we accept the iterative conception of sets, then we are violating the Domain Principle. For instance, “V” is showing an open universe of sets arranged with sets of sets that continue to get larger. This returns the form of a big “V” implying that we have an open potential of sets that are ordered such that we can not group them into a completed totality containing all sets. We have already seen this problem with Russell’s set containing all sets which is something we want to avoid. Therefore, if we truly want the truth, we must accept that this contradiction will arise when discussing sets of totality. Furthermore, if we accept this, then

we can truly look under the hood of the problem.

6

[**Extra Credit**]. In 1966, in the book in which he popularized his methods of forcing, P. J. Cohen wrote: “A point of view which the author feels may eventually come to be accepted is that the CH [Continuum Hypothesis] is obviously false. The main reason one accepts the Axiom of Infinity is probably that we feel it absurd to think that the process of adding only one set at a time can exhaust the entire universe . . . Now $\aleph_1$ is the set of countable ordinals and this is merely a special and the simplest way of generating a higher cardinal. The set C [the set of Reals] is, in contrast, generated by a totally new and more powerful principle, namely the Power Set Axiom. It is unreasonable to expect that any description of a larger cardinal which attempts to build up that cardinal from ideas deriving from the Replacement Axiom can ever reach C . . . This point of view regards C as an incredibly rich set given to us by one bold new axiom, which can never be approached by any piecemeal process of construction.” (Cohen, *Set Theory and the Continuum Hypothesis*, p. 151). In what sense is the Power Set Axiom “more powerful” than the Axioms of Infinity and Replacement, and what reason is there to think that it allows us to generate sets that cannot be generated by any “piecemeal process of construction”? Do you agree or disagree with the general viewpoint that Cohen expresses here? Why?

The Power Set Axiom is more powerful than the Axiom of Infinity and Replacement because it gives us the ability to generate larger sets than that of the Axiom of Infinity and Replacement. The Power Set Axiom is a diagonalizer which means that it has the ability to take us out of the current set and generate a new set such that it is not a member of the old set, but still has the correct form. Thus, we can generate sets like the set of all Real numbers. The Axiom of Infinity allows us to generate an infinite set of out nothing that is useable for other axioms; however, the infinite set generated does not have a specific infinity but does have the cardinality of $\aleph_1$. The Axiom of Replacement generates sets of lower cardinalities, but this means that the cardinality of the elements is constructible meaning we can't increase cardinality.

Furthermore, The Power Set Axiom allows us to create sets that cannot be generated by a “piecemeal process of construction” because it diagonalizes out a new and larger set from uncountable sets. For instance, the proof we did on the first day where we diagonalized a new set of Natural numbers that allowed us to show that the set of real numbers is equivalent to the set of Natural numbers. We did not perform a series of steps to get an answer, instead we performed one step (circling numbers in a diagonal line) to get a new and larger set. This happens because when we diagonalize a set, we take the complement of each element in the set.

I do not agree with Cohen because I believe the Continuum Hypothesis is real and it is consistent with ZFC.